

ADAM-TPU: Rotation as a Shared Hardware Primitive for Analog LLM Inference

A Complex-Polar Spectral Substrate Unifying Rotary Embeddings, Geometric-Algebra Rotations, and Rotation-Based KV

Quantization

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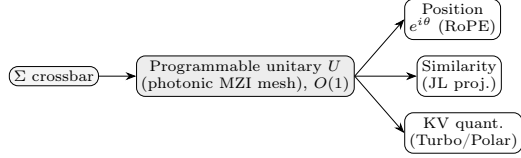


Figure 1: One programmable rotation, time-shared across positional encoding, similarity, and quantization; the crossbar supplies the diagonal scaling Σ .

Abstract

We argue that a single physical primitive—a programmable *unitary rotation*—underlies three otherwise-separate problems in LLM serving: rotary positional encoding, which is complex multiplication on $\mathbb{C}^{d/2}$ [5]; inner-product-preserving similarity, a Johnson–Lindenstrauss projection [4]; and near-optimal KV-cache quantization, which works by randomly rotating activations so coordinates become near-independent and admit optimal per-coordinate scalar quantization [1,2,3]. Coherent photonic interferometer meshes realize arbitrary unitaries in one $O(1)$ settling step [10,11], and resistive crossbars realize the diagonal scaling by Ohm–Kirchhoff summation [12]. ADAM-TPU proposes a complex-polar analog substrate in which one rotation engine is time-shared across position, attention, and compression, paired with a recurrent low-rank state for context [6,7] and a measured-boot attestation layer for integrity. The key consequence: the limited precision of analog hardware (~ 6 – 8 bits) is *not* fatal for the memory path, because rotation-based quantization is quality-neutral at 2.5–3.5 bits/channel [1]—inside that budget—and the rotation is exactly what the optics provide for free.

1 Thesis: one rotation, three roles

Normalized embeddings live on a sphere, so the natural number system is polar/complex, where a rotation is a phase addition,

$$z_1 z_2 = (r_1 r_2) e^{i(\phi_1 + \phi_2)}, \quad (1)$$

turning a multiply into an add (the CORDIC regime). This is not metaphor: rotary position embedding (RoPE) is exactly multiplication by $e^{i\theta}$ on coordinate pairs [5], so the dominant positional scheme in modern LLMs *already* is complex rotation. The primitive recurs twice more. (i) Random rotation is the engine of state-of-the-art KV quantization: rotating activations renders coordinates near-Gaussian and near-independent, so independent optimal scalar quantizers reach near information-theoretic distortion [1,3], and POLARQUANT does so in an explicitly polar parametrization [2]. (ii) Inner products survive random projection (Johnson–Lindenstrauss); a 1-bit sign-projection already yields *unbiased* inner-product estimates [4]. A programmable unitary is thus a universal currency, which ADAM-TPU mints in physics rather than spending digital multiplies on (Fig. 1).

2 Substrate, compression, integrity

Substrate. A mesh of Mach–Zehnder interferometers realizes any unitary $U, V^\top$ by construction [10,11]; a crossbar realizes the diagonal Σ and

the MVM in one settling step, $I_j = \sum_i V_i G_{ij}$ [12]. Signed/complex conductances $G = G_{\Re} + iG_{\Im}$ are realizable with paired cells, and complex CORDIC supplies the angle datapath. Weights are deployed factored, $W = U\Sigma V^\top = \sum_k \sigma_k u_k v_k^\top$: optics do rotations, the crossbar scaling, and rank K is a runtime energy knob (gate the tail $k > K$; optimal by Eckart–Young). A VLIW front-end fixes the schedule at compile time, sparing dynamic-scheduler silicon. *Two honest caveats*: a crossbar *applies* a known factorization—it does not *compute* the SVD; in-loop analog eigensolving is shown only at small scale [13], so offline factorization is assumed. And arbitrary weights are *not* Hermitian, so the spectral theorem’s real-eigenvalue guarantee does not apply directly—SVD (any matrix; $W^\top W$ for the real spectrum) is the right general tool, not a forced Hermitian projection.

Compute-side compression. Context is a fixed-size recurrent low-rank state,

$$A_{t+1} = \lambda A_t + \mathbf{x}_t \mathbf{y}_t^\top, \quad (2)$$

the linear-attention / state-space recurrence [6,7], trading $O(L^2)$ KV growth for a constant state.

Storage-side compression. What remains is quantized *after rotation*: TURBOQUANT attains quality-neutral long-context retrieval at 3.5 bits and $> 5\times$ compression, matching full precision on needle-in-a-haystack at $4\times$ [1]. Here the substrate *improves* on the software method—TurboQuant’s rotation costs a GPU matmul but is *free* on a photonic mesh, and its 2.5–3.5-bit target sits inside the analog noise floor, so analog precision (the usual objection) is a non-issue for memory. Shannon’s source-coding bound caps distortion [14]; the design should *target* it, not beat it. FFT token mixing (FNET) [8] and geometric-algebra rotors [9] are further unitary, hardware-cheap options generalizing (1) without gimbal lock.

Integrity (bounded). A PUF-rooted measured boot attests the loaded configuration against the published digest; control-flow integrity guards the run; a self-timed layout makes physical edits perturb handshake timing (tamper-evidence, not -proofing). This certifies *which* configuration ran—not that outputs are correct—from standard primitives, not a new cryptographic claim.

3 Socratic Tail: open problems & license

(1) **Apply $\neq$ solve**: physical decomposition at LLM scale is unsolved; [13] works only at small dimension—the central risk. (2) **Analog noise**: resolved for the quantized *memory* path, open for high-precision *compute*/audit. (3) **Conversion wall**: ADC/DAC energy, not the MAC, dominates—the analog win is latency, not free energy. (4) **Drift & endurance**: memristors drift thermally with finite write endurance, hindering calibration and on-device training. (5) **Lossy floor**: high-entropy sequences exceed truncation rank and the fixed state; the Shannon bound [14] is fundamental. (6) **Hermitian-ization is not free**. (7) **Integrity is bounded**; photonic alignment and cold-boot calibration add yield and attack surface. *Released under the S.V.E. Meta-License v4.0 (share-alike); the limitations are part of the contribution, not footnotes to it.*

References. [1] Zandieh et al., “TurboQuant,” arXiv:2504.19874, 2025. [2] Han et al., “PolarQuant,” arXiv:2502.02617, 2025. [3] Ashkboos et al., “QuaRot,” arXiv:2404.00456, 2024. [4] Zandieh et al., “QJL: 1-bit Quantized JL Transform,” arXiv:2406.03482, 2024. [5] Su et al., “RoFormer (RoPE),” arXiv:2104.09864, 2021. [6] Katharopoulos et al., “Transformers are RNNs (linear attention),” ICML 2020. [7] Gu & Dao, “Mamba,” arXiv:2312.00752, 2023. [8] Lee-Thorp et al., “FNet,” arXiv:2105.03824, 2021. [9] Brandstetter et al., “Clifford Neural Layers,” ICLR 2023 (arXiv:2209.04934). [10] Shen et al., Nature Photonics 11:441, 2017. [11] Clements et al., Optica 3:1460, 2016 (cf. Reck et al., PRL 73:58, 1994). [12] Sebastian et al., Nature Nanotechnology 15:529, 2020. [13] Sun et al., PNAS 116:4123, 2019. [14] Shannon, Bell Syst. Tech. J., 1948.

*Email: artiomkovnatsky@pm.me. Repo: <https://codeberg.org/skovnats/SVE-Systemic-Verification-Engineering>. Released under the S.V.E. Meta-License v4.0 (share-alike). Position/architecture paper: nothing here is a fabricated-and-measured result; cited numbers are from the literature, and the open problems (§3) are part of the contribution.